Perioperative Management of Antiplatelet Therapy

Objective

To provide guidance to clinicians, based on the American College of Chest Physicians Clinical Practice Guideline (2022) and the ESC Guidelines on cardiovascular assessment and management of patients undergoing non cardiac surgery (2022) for the perioperative management of patients on antiplatelet therapy who require non-cardiac or cardiac surgery.

Background

Perioperative management can be challenging in patients who are receiving acetylsalicylic acid (ASA), a P2Y12 inhibitor (clopidogrel, prasugrel, ticagrelor), or both ASA and a P2Y12 inhibitor. Clinicians should balance the risks of major adverse cardiovascular events associated with interrupting these therapies against the risk of bleeding from continuing these therapies in the perioperative period. Additionally, other factors including the pharmacokinetic actions of antiplatelet drugs and the optimal timing of surgery in patients with coronary stenting should be considered. The latter group of patients requires special consideration due to the increased risks and significant mortality of stent thrombosis.

Risk Stratification for Perioperative Thrombosis and Bleeding

Risk stratification for thrombosis and bleeding is largely empiric in patients who are receiving antiplatelet therapy.

Patients considered at high risk for perioperative cardiovascular events include those with a recent (i.e. within one year) implanted bare metal stents (BMS) or drug-eluting stents (DES) or a recent myocardial infarction (MI), systemic embolism, or stroke.

Patients at low risk for perioperative cardiovascular events are those taking antiplatelet therapy for primary prevention of MI or stroke. Clinicians should balance these risks against the associated risks of perioperative bleeding (**Table 1**).

Table 1: Risk of Perioperative Bleeding

Management in Patients Having a Low/Very Low Bleed Risk Procedure

Patients undergoing arthrocentesis, minor dental (extraction, root canal), eye (cataract) or skin (biopsy, skin cancer excision) procedures, as well as low bleeding risk diagnostic procedures, can continue ASA without interruption. It is reasonable to continue a P2Y12 inhibitor around the procedure if taken as monotherapy. If patients are taking dual antiplatelet therapy, management should be individualized based on assessment of thrombotic vs. bleeding risk.

Management in Patients without Coronary Stents Undergoing Elective or Non-Urgent Noncardiac Surgery

The Perioperative Ischemic Evaluation (POISE) 2 randomized control trial evaluated the perioperative antiplatelet drug management in noncardiac surgery. This placebo-controlled trial assessed immediate pre-operative ASA initiation (200 mg) or ASA continuation in 10,010 patients with known or at risk for coronary artery disease. It demonstrated that continuing/initiating ASA did not reduce the risk for non-fatal MI or death (7.0% vs 7.1%; RR 0.99; 95% CI: 0.86-1.2), a finding observed in both initiation and continuation strata but increased the risk for major bleeding (4.6% vs 3.8%; RR 1.2; 95% CI: 1.01-1.50).

The PEP trial involved 17,444 patients who required hip fracture or hip/knee joint replacement. Patients were randomized to receive ASA or placebo started preoperatively and continued for 35 days. ASA use was associated with a decreased risk for VTE (RR 0.71; 95% CI: 0.54-0.94) and did not affect the risk for MI (RR 1.57; 95% CI: 0.93-2.65) or stroke (RR 1.13; 95% CI: 0.69-1.85) but conferred an increased risk for major bleeding (2.9% vs 2.4%; P = 0.04).

The decision to continue ASA preoperatively should take bleeding risk of the surgical procedure into account. Clinical situations of when ASA should be continued perioperatively in patients without coronar

undergoing elective or non-urgent noncardiac surgery include:

- Patients undergoing carotid endarterectomy or those with recent acute coronary syndrome
- Patients who are undergoing surgeries with low to moderate risk of perioperative bleeding (Table 1)
- Perioperative ASA continuation might also be reasonable for surgical interventions to prevent local thrombosis (e.g. lower extremity bypass or arterial aneurysm repair).

Initiating ASA before surgery in previous non-users of ASA to reduce perioperative cardiovascular events is not recommended.

ASA interruption can be considered in patients without coronary stents or a recent thrombotic event undergoing a non-cardiac surgery associated with a high-bleeding risk (Table 1). If ASA interruption is adopted, we suggest interruption for 7 days. After surgery, we suggest resumption of ASA when the risk of bleeding related to surgery has passed, typically within 24 hours.

Data regarding the perioperative management of patients who are undergoing noncardiac surgery and are on P2Y12 inhibitor monotherapy is lacking. Consultation with a specialist is advised, as situations for P2Y12 inhibitor monotherapy is variable (e.g. a de-escalation strategy after an ACS, longer-term secondary cardiovascular event prevention, peripheral arterial disease, or ASA intolerance) and thrombotic risk can vary considerably. Management may also vary from switching to ASA perioperatively or continuing or interrupting the P2Y12 inhibitor perioperatively. If the decision is made to interrupt the P2Y12 inhibitor, the P2Y12 inhibitor should be restarted when the risk of bleeding related to the surgery has passed, typically within 24 hours postoperatively. The timing of interruption of these medications is generally

- 7 days preoperatively for prasugrel
- 5 days preoperatively for clopidgrel
- 3-5 days preoperatively for ticagrelor

Management of Patients with Coronary Stents Undergoing Elective or Non-Urgent Noncardiac Surgery

A sub-study of the POISE-2 trial involving 470 patients with previous percutaneous coronary intervention (PCI) and cardiac stents revealed that for every 1,000 patients with prior PCI, perioperative ASA will prevent 59 myocardial infarctions but cause 8 major bleeds. It is, therefore, viewed that in patients with prior PCI undergoing noncardiac surgery, perioperative ASA may be more likely to benefit than harm patients.

A retrospective study of 20,590 patients with coronary stents who underwent non cardiac surgery, demonstrated that the risk for adverse cardiovascular outcomes appeared highest in the initial 6 weeks after stent placement (8%-10%), appeared to plateau at 6 months (1%-2%), and remain stable at 24 months.

Clinicians must consider the timing of surgery and perioperative dual antiplatelet (DAPT) management in patients being treated with DAPT after PCI with a BMS or DES, as well as the thrombotic risk (Table 1). In patients:

- With PCI ≥ 12 months ago à continue ASA perioperatively if the bleeding risk allows.
- With an elective PCI
    - Ideally delay elective noncardiac surgery and continue dual antiplatelet therapy until 6 months after elective PCI. After 6 months, perioperative management includes continuing ASA perioperatively and perioperative interruption of the P2Y12 inhibitor (if still on dual antiplatelet therapy).
    - Delay time-sensitive, noncardiac surgery and continue dual antiplatelet therapy for a minimum of 1 month. After 1 month, perioperative management includes continuing ASA perioperatively and perioperative interruption of the P2Y12 inhibitor.

- With a PCI for ACS
  - Ideally delay elective noncardiac surgery and continue dual antiplatelet therapy until 12
    After 12 months, perioperative management includes continuing ASA perioperatively and perioperative interruption of the P2Y12 inhibitor (if still on dual antiplatelet therapy)
  - Delay time-sensitive, noncardiac surgery and continue dual antiplatelet therapy for 3 months should be considered. Perioperative management includes continuing ASA perioperatively and perioperative interruption of the P2Y12 inhibitor.
  - Specialist consultation and a multidisciplinary discussion is advised in situations when surgery is required within 3 months of a PCI for ACS.
- With other high-risk factors for perioperative stent thrombosis (e.g. past stent thrombosis, left main PCI) and are still on dual antiplatelet therapy, specialist consultation is advised.

Table 2: Timing of Surgery in PCI Patients with Stents

Management of Patients Requiring Elective or Semi-Urgent CABG after Acute Coronary Syndrome (ACS)(Table 3)

Coronary artery bypass grafting is associated with a high risk of bleeding with potentially significant consequences (i.e., cardiac tamponade, death). An interdisciplinary assessment of the risks of coronary thrombotic complications and risk of perioperative bleeding should be performed with the surgeon, interventional cardiologist, attending physician / cardiologist and the patient.

Table 3: Management of antiplatelet therapy in patients undergoing CABG

Other Relevant

- Acetylsalicylic Acid (ASA®)
- Clopidogrel (Plavix®)
- Prasugrel (Effient®)
- Ticagrelor (Brilinta®)

s clinical guides remain current and reliable. Your support helps protect the quality and continuity of these resources for clinicians nationwide.

References

Devereaux PJ, et al. Aspirin in patients undergoing noncardiac surgery. New Engl J Med 2014; 370: 1494-1503.

Douketis JD, et al. Perioperative Management of Antithrombotic Therapy: An American College of Chest Physicians Clinical Practice Guideline. *Chest*. 2022;162(5):e207-e243.

Douketis J, Spyropoulos A. Perioperative management of anticoagulant and antiplatelet therapy. NEJM Evid 2023;2(6).

Graham M et al. Aspirin in patients with previous percutaneous coronary intervention undergoing noncardiac surgery. Ann Intern Med 2018;168(4):237-244.

Halvorsen S, et al. 2022 ESC Guidelines on cardiovascular assessment and management of patients undergoing non-cardiac surgery. *Eur Heart J*. 2022;43(39):3826-3924. doi:10.1093/eurheartj/ehac270

Holcomb CN, et al. The incremental risk of noncardiac surgery on adverse cardiac events following coronary stenting. J Am Coll Cardiol. 2014;64(25):2730-2739.

Mehta SR, et al. 2018 Canadian Cardiovascular Society/Canadian Association of Interventiona
update of the guidelines for the use of antiplatelet therapy. Can J Cardiol 2018;34(3):214-233.

Prevention of pulmonary embolism and deep vein thrombosis with low dose aspirin: Pulmonary Embolism Prevention (PEP) trial. Lancet. 2000;355(9212):1295-1302.

Date of Version: 21April2025

Please note that the information contained herein is not to be interpreted as an alternative to medical advice from your doctor or other professional healthcare provider. If you have any specific questions about any medical matter, you should consult your doctor or other professional healthcare providers, and as such you should never delay seeking medical advice, disregard medical advice or discontinue medical treatment because of the information contained herein.